LABORATORY OBSERVATION

Course	Full Stack Web Development
Course Code	23CM4121
Observation Task No.	4
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1. TITLE

Design and Implementation of a Daily Task Manager Using HTML, CSS and JavaScript

2. AIM

To develop an interactive Daily Task Manager that allows users to add tasks dynamically, mark tasks as completed, and remove tasks using JavaScript DOM manipulation.

3. OBJECTIVE

1. To create a simple task management interface using HTML.
2. To style the task manager using CSS.
3. To accept task information through an input field.
4. To dynamically create list items using JavaScript.
5. To mark a task as completed using classList.toggle().
6. To remove a task using parentElement.remove().
7. To understand event handling and DOM manipulation.

4. THEORY

JavaScript can manipulate the Document Object Model (DOM) to create and modify webpage elements. The getElementById() method is used to access HTML elements. The createElement() method creates new elements dynamically, while appendChild() inserts them into the document. The classList.toggle() method adds or removes a CSS class, which is used here to mark a task as completed. The parentElement.remove() method removes the selected task from the webpage.

Application Flow: Enter Task → Validate Input → Create List Item → Add to Task List → Mark Complete / Remove Task

5. ALGORITHM / PROCEDURE

1. Create an HTML document titled 'Daily Task Manager'.
2. Design a centered task-management container using CSS.
3. Create an input field to accept a new task.
4. Create an Add Task button that calls createTask().
5. Read and trim the entered task value.
6. If the input is empty, stop the function.
7. Create a new li element using createElement().
8. Insert a task span and Remove button into the new list item.
9. Append the list item to the task list.
10. Clear the input field after adding the task.

11. Use completeTask() to toggle the completed CSS class.
12. Use deleteTask() to remove the selected list item.
13. Open the webpage in a browser and verify all operations.

6. PROGRAM

task_manager.html

```
<!DOCTYPE html>
<html>

<head>

<title>Daily Task Manager</title>

<style>

body {
    font-family: Verdana, sans-serif;
    background: #fff7ed;
    display: flex;
    justify-content: center;
    padding: 60px;
}

.container {
    background: white;
    padding: 35px;
    width: 520px;
    border-radius: 12px;
    box-shadow: 0 6px 18px #bbb;
}

h2 {
    text-align: center;
    color: #ea580c;
    font-size: 28px;
}

.input-area {
    display: flex;
    gap: 12px;
}

input {
    flex: 1;
    padding: 14px;
    font-size: 17px;
    border: 1px solid #bbb;
    border-radius: 7px;
}

button {
    background: #ea580c;
    color: white;
    border: none;
    padding: 14px 20px;
    font-size: 16px;
    border-radius: 7px;
    cursor: pointer;
}

button:hover {
    background: #c2410c;
}

ol {
    padding-left: 25px;
}

li {
    background: #fff7ed;
    margin: 12px 0;
    padding: 14px;
    font-size: 17px;
    border-radius: 7px;
    display: flex;
    justify-content: space-between;
    align-items: center;
}

.completed {
    text-decoration: line-through;
    color: #999;
}

.remove {
    background: #dc2626;
    padding: 7px 11px;
    font-size: 13px;
}

.remove:hover {
    background: #991b1b;
}

</style>

</head>
```

```
<body>

<div class="container">

<h2>■ Daily Task Manager</h2>

<div class="input-area">

<input id="taskInput" placeholder="Write a new task">

<button onclick="createTask()">Add Task</button>

</div>

<ol id="taskList"></ol>

</div>

<script>

function createTask() {

    let taskBox = document.getElementById("taskInput");

    let taskText = taskBox.value.trim();

    if (taskText === "") {
        return;
    }

    let newTask = document.createElement("li");

    newTask.innerHTML = `
        <span onclick="completeTask(this)">
            ${taskText}
        </span>

        <button class="remove"
        onclick="deleteTask(this)">
            Remove
        </button>
    `;

    document.getElementById("taskList").appendChild(newTask);

    taskBox.value = "";

}

function completeTask(element) {

    element.classList.toggle("completed");

}

function deleteTask(button) {

    button.parentElement.remove();

}

</script>

</body>

</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
getElementById()	Selects an HTML element using its unique ID.
value.trim()	Gets the input value and removes unnecessary spaces.
createElement()	Creates a new HTML list item dynamically.
innerHTML	Adds the task text and Remove button inside the list item.
appendChild()	Adds the newly created task to the task list.
classList.toggle()	Adds or removes the completed class to mark a task complete.
parentElement.remove()	Removes the selected task from the webpage.
onclick	Calls a JavaScript function when a button or task is clicked.
.completed	CSS class that applies a line-through style to completed tasks.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Enter 'Complete assignment'	Task is accepted	Accepted	Pass
TC-02	Click Add Task	New task appears in list	Task added	Pass
TC-03	Click task text	Task gets line-through style	Marked completed	Pass
TC-04	Click Remove	Selected task disappears	Task removed	Pass
TC-05	Click Add Task with empty input	No task should be added	No task added	Pass

9. OUTPUT

Output: A Daily Task Manager interface is displayed with an input field and Add Task button. Entered tasks are dynamically displayed as list items. Clicking a task marks it as completed, while clicking Remove deletes the task.

■ Daily Task Manager

Write a new task [Add Task]

1. Complete assignment [Remove]
2. Study JavaScript [Remove]

Completed task example: ~~Submit lab record~~

10. RESULT

Thus, the Daily Task Manager was successfully developed using HTML, CSS, and JavaScript. The program successfully accepts tasks, dynamically adds them to the list, marks tasks as completed using CSS class manipulation, and removes tasks from the DOM.